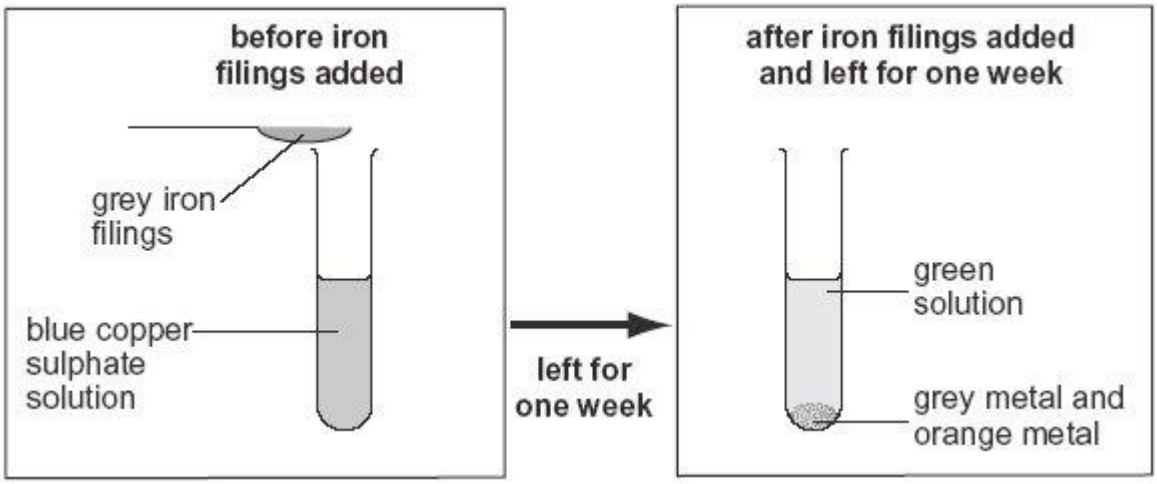
Q1.

Joanne added iron filings to copper sulphate solution.
She observed the reaction after one week.



(a) What evidence in the diagrams shows that a chemical reaction has taken place?

.....

1 mark

(b) The reaction between iron and copper sulphate is a **displacement** reaction.

(i) Give the name of the orange metal visible after one week.

.....

1 mark

(ii) What is the name of the compound formed in this reaction?

.....

1 mark

(iii) Joanne poured the green solution into another test tube. She added some copper pieces to the solution.

Will a displacement reaction occur?

yes ☐ no ☐

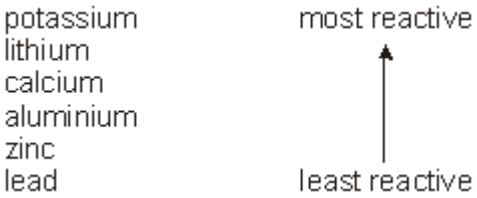
Explain your answer.

.....

.....

1 mark

(c) Part of the reactivity series of metals is shown below.



Use the information above.
Which **two** metals would react with aluminium nitrate in a displacement reaction?

Tick the **two** correct boxes.

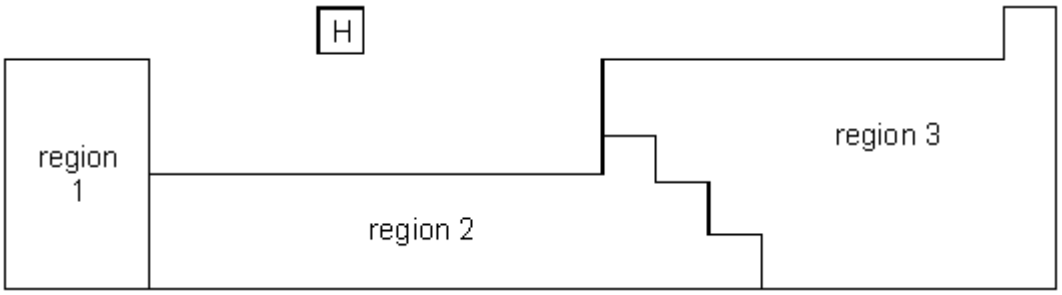
calcium ☐ potassium ☐

zinc ☐ lead ☐

1 mark
maximum 5 marks

Q2.

The diagram shows an outline of part of the Periodic Table of Elements.



(a) What is the name of the element with the symbol H?

……

1 mark

(b) In which regions of the Periodic Table are the following types of element found?

(i) non-metals (such as oxygen and chlorine);

region ……

1 mark

(ii) very reactive metals (such as sodium and potassium);

region ……

1 mark

(iii) less reactive metals (such as copper and zinc).

Region ……

1 mark

(c) Why is copper sulphate **not** found in the Periodic Table?

……

……

1 mark

(d) An iron nail is placed into some blue copper sulphate solution.
A reaction takes place between the iron and the copper sulphate.

(i) Complete the word equation for the reaction.

iron + copper sulphate → …… +……

1 mark

(ii) Describe **one** change you would see on the surface of the nail.

……

……

1 mark

Maximum 7 marks

Q1.

(a) any **one** from

- there is a colour change

  *accept 'it goes green **or** orange'*
  'the colour' is insufficient

- a new metal is formed

  accept 'the iron filings change colour'

1 (L5)

(b) (i) copper

accept 'Cu'

1 (L5)

(ii) iron sulphate

accept '$FeSO_4$'

1 (L6)

(iii) • no ✓

any **one** from

- iron is more reactive than copper

  accept 'iron is higher on the reactivity series'

- copper is less reactive than iron

  accept 'copper does not displace iron'

  ***both** an indication that the reaction does not happen and the explanation are required for the mark*

1 (L6)

(c) • calcium ✓
potassium ✓

if more than two boxes are ticked, award no mark
***both** answers are required for the mark*

1 (L6)

[5]

Q2.

(a) hydrogen

1 (L6)

(b) (i) region 3

1 (L6)

(ii) region 1

1 (L6)

(iii) region 2

1 (L6)

(c) any **one** from

- it is a compound
- it is not an element
- it is made up of more than one element

*do **not** accept 'it is **not** a single substance'*

1 (L5)

(d) (i) copper + iron sulphate

answers may be in either order
***both** are required for the mark*

1 (L6)

(ii) the nail becomes brown **or** pink **or** copper coloured

accept 'it is covered in copper'
accept 'it is rust coloured'
*do **not** accept 'it goes rusty'*

1 (L6)

[7]